EDF with constrained deadlines: bounding complexity (2/2)

$= \sum_i (T_i - D_i)\frac{C_i}{T_i} + \sum_i t\frac{C_i}{T_i} =$

4. $\text{dbf}(t) = \sum_{i=1}^{n} \left\lfloor \frac{t + T_i - D_i}{T_i} \right\rfloor C_i \leq \sum_{i=1}^{n} \frac{t + T_i - D_i}{T_i} C_i = \sum_{i=1}^{n} (T_i - D_i)U_i + tU \Rightarrow$

   $\Rightarrow G(0,t) := \sum_{i=1}^{n} (T_i - D_i)U_i + tU$ is an increasing function with slope $U \Rightarrow$

   $\Rightarrow$ if $U < 1$ then $\exists\, t^\star \mid G(0,t^\star) = t^\star$ where $t^\star = \sum_{i=1}^{n} (T_i - D_i)U_i/(1-U) \Rightarrow$

   $= y(t^*)$ RISOLVENDO

   $\Rightarrow \text{dbf}(t) \leq G(0,t) \leq t \;\; \forall\, t \geq t^\star \Rightarrow$ Verify the criterion only for $t < t^\star$

   (note that the task set considered in figure below is NOT schedulable)

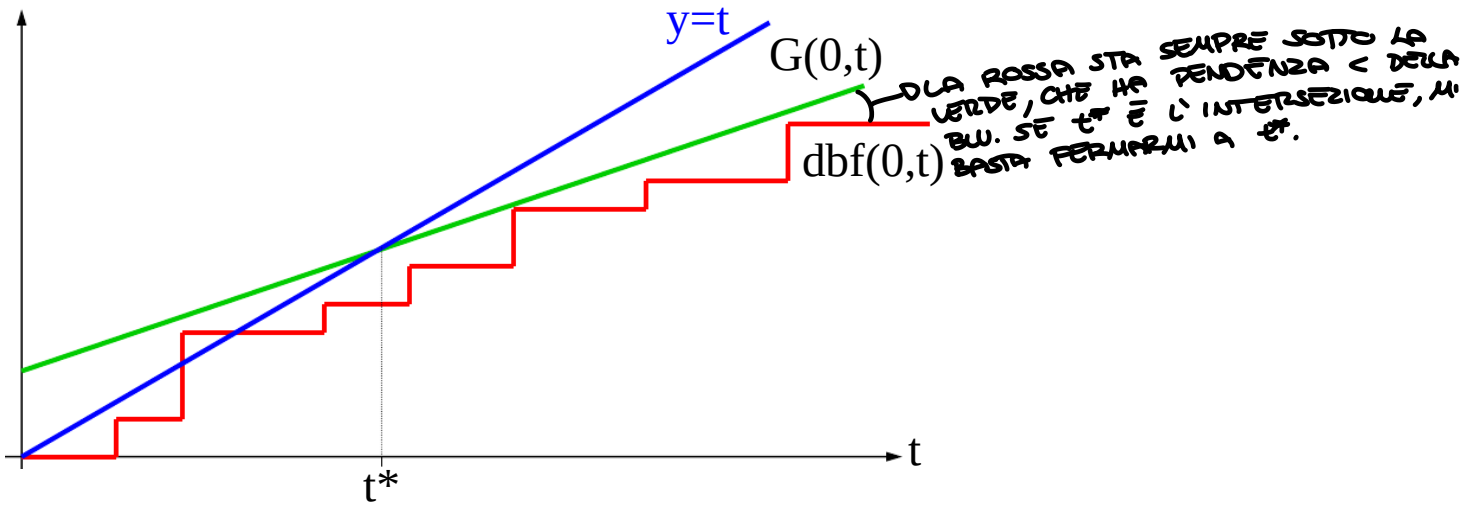

Image adapted from Prof. G. Buttazzo, Scuola Superiore Sant'Anna, Pisa